Linear algebra: Exercise chapter 11

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1. Find the singular value decomposition of the following matrices.

$$A = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Answer.

Matrix A.

Step 1: Compute A^TA and AA^T:

$$A^TA = \begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix},$$
$$AA^T = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix}.$$

Step 2: Eigenvalues and Singular Values:
Eigenvalues of A^TA:

$$\det(A^TA - \lambda I) = \begin{vmatrix} 5-\lambda & 5 \\ 5 & 5-\lambda \end{vmatrix} = (5-\lambda)^2 - 25 = 0 \Longrightarrow \lambda_1 = 10, \quad \lambda_2 = 0.$$

Singular values:

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{10}, \quad \sigma_2 = \sqrt{\lambda_2} = 0.$$

Step 3: Eigenvectors of AA^T:
For $\lambda = 10$:

$$(AA^T - 10I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} -2 & -4 \\ -4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \Rightarrow \quad x_1 = -2x_2 \quad \Rightarrow \quad u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

For $\lambda = 0$:

$$(AA^T - 0I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \Rightarrow \quad x_2 = 2x_1 \quad \Rightarrow \quad u_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Step 4: Eigenvectors of A^TA:
For $\lambda = 10$:

$$(A^TA - 10I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \Rightarrow \quad x_1 = x_2 \quad \Rightarrow \quad v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda = 0$:

$$(A^TA - 0I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad \Rightarrow \quad x_1 = -x_2 \quad \Rightarrow \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Step 5: Construct U, Σ, V^T:

$$U = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{bmatrix}, \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

SVD: $A = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}.$

Matrix B.

Step 1: Compute B^TB and BB^T:

$$B^TB = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 6 & 12 \\ 12 & 24 \end{bmatrix}.$$

$$BB^T = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 10 & -5 \\ 10 & 20 & -10 \\ -5 & -10 & 5 \end{bmatrix}.$$

Step 2: Eigenvalues and Singular Values of B^TB:
Eigenvalues of B^TB:

$$\det(B^TB - \lambda I) = \begin{vmatrix} 6-\lambda & 12 \\ 12 & 24-\lambda \end{vmatrix} = (6-\lambda)(24-\lambda) - 144 = 0 \Longrightarrow \lambda_1 = 30, \quad \lambda_2 = 0.$$

Singular Values:

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{30}, \quad \sigma_2 = \sqrt{\lambda_2} = 0.$$

Step 3: Eigenvectors of BB^T:
For $\lambda = 30$:

$$(BB^T - 30I)x = 0 \Rightarrow \begin{bmatrix} -25 & 10 & -5 \\ 10 & -10 & -10 \\ -5 & -10 & -25 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{cases} -5x_1 + 2x_2 - x_3 = 0 \\ x_1 - x_2 - x_3 = 0 \\ x_1 + 2x_2 + 5x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_2 = 2x_1 \\ x_3 = -x_1 \end{cases} \Rightarrow u_1 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix}.$$

For $\lambda = 0$:

$$(BB^T - 0I)x = 0 \Rightarrow \begin{bmatrix} 5 & 10 & -5 \\ 10 & 20 & -10 \\ -5 & -10 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow x_1 = -2x_2 + x_3 \Rightarrow u_2 = \begin{bmatrix} \frac{-2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}, \quad u_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Step 4: Eigenvectors of B^TB:
For $\lambda = 30$:

$$(B^TB - 30I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} -24 & 12 \\ 12 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_2 = 2x_1 \Rightarrow \quad v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

For $\lambda = 0$:

$$(B^TB - 0I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 6 & 12 \\ 12 & 24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_1 = 2x_2 \Rightarrow \quad v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Step 5: Construct U, Σ, V^T:

$$U = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}.$$

SVD of B:

$$B = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{5}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & 0 \\ \frac{-1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{30} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}.$$

Matrix C.

Step 1: Compute C^TC and CC^T:

$$C^TC = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix},$$

$$CC^T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Step 2: Eigenvalues and Singular Values:
Eigenvalues of C^TC:

$$\det(C^TC - \lambda I) = \begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)\big((1-\lambda)^2 - 1\big) = 0 \Longrightarrow \lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = 0.$$

Singular Values:

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{2}, \quad \sigma_2 = \sqrt{\lambda_2} = 1, \quad \sigma_3 = \sqrt{\lambda_3} = 0.$$

Step 3: Eigenvectors of CC^T:
For $\lambda = 2$:

$$(CC^T - 2I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \;\Rightarrow\; x_2 = 0 \;\Rightarrow\; u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

For $\lambda = 1$:

$$(CC^T - 1I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \;\Rightarrow\; x_1 = 0 \;\Rightarrow\; u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Step 4: Eigenvectors of C^TC:
For $\lambda = 2$:

$$(C^TC - 2I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \;\Rightarrow\; \begin{cases} x_1 = -x_3 \\ x_2 = 0 \end{cases} \;\Rightarrow\; v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

For $\lambda = 1$:

$$(C^TC - 1I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \;\Rightarrow\; \begin{cases} x_3 = 0 \\ x_1 = 0 \end{cases} \;\Rightarrow\; v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

For $\lambda = 0$:

$$(C^TC - 0I)x = 0 \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \;\Rightarrow\; \begin{cases} x_1 = x_3 \\ x_2 = 0 \end{cases} \;\Rightarrow\; v_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Step 5: Construct U, Σ, V^T:

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

SVD of C:

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

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2. For the matrix B in the previous exercise, find a basis for each matrix four space.
Answer.

$$C(B) = \text{span}\left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right), \quad N(B^T) = \text{span}\left(\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right), \quad C(B^T) = \text{span}\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right), \quad N(B) = \text{span}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right).$$

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3. Consider the following matrix: $A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$.

   (a) Both the columnspace and the rowspace are 1-dimensional. Find a basis for both spaces.
   (b) Why is σ_2 zero (without calculating it)?
   (c) Without solving systems $Ax = 0$ and $A^Tx = 0$, find a basis for $N(A)$ and $N(A^T)$.

Answer.
(a):

$$C(A) = \text{span}\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right), \quad C(A^T) = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right).$$

(b): $A \in \mathbb{R}^{2\times 2}$ and $\dim(A) = \dim(C(A)) = \dim(C(A^T)) = 1$. Thus, $\dim(N(A)) = \dim(N(A^T)) = 1$. Consequently, A^TA has a zero eigenvalue, implying that A has a zero singular value.

(c): Since $C(A)^{\perp} = N(A^T)$ and $\begin{bmatrix} -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ are orthogonal, thus

$$N(A^T) = \text{span}\left(\begin{bmatrix} -3 \\ 1 \end{bmatrix}\right).$$

Furthermore, since $C(A^T)^{\perp} = N(A)$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are orthogonal, thus

$$N(A) = \text{span}\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right).$$

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4. Assume that $A \in \mathbb{S}^{n\times n}$ is an invertible matrix.
   (a) What is the singular value decomposition of A^{-1}?
   (b) What are the singular values of A^{-1}?

   Answer.
   (a): Since $A \in \mathbb{S}^{n\times n}$, the singular value decomposition (SVD) of A is given by: $A = U\Sigma V^T = Q\Sigma Q^T$, where $Q = U = V$, $QQ^T = I$, and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$ is a diagonal matrix containing the singular values of A ($\sigma_i > 0$, since A is invertible). The inverse of A can be expressed as:

$$A^{-1} = (Q\Sigma Q^T)^{-1} = (Q^T)^{-1}\Sigma^{-1}Q^{-1} = Q\Sigma^{-1}Q^T,$$

   where $\Sigma^{-1} = \text{diag}\left(\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \ldots, \frac{1}{\sigma_n}\right)$.
   (b): The singular values of A are $\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \ldots, \frac{1}{\sigma_n}$.

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5. Why doesn't the singular value decomposition of $A + I$ just use $\Sigma + I$?
   Answer. Assume that $A \in \mathbb{R}^{n\times n}$ and σ is a singular value of A. Then, $\lambda = \sigma^2$ is an eigenvalue of A^TA. Now, if $\sigma + 1$ is a singular value of $A + I$, it implies that $(\sigma+1)^2 = \sigma^2 + 2\sigma + 1$ is eigenvalue of $(A+I)^T(A+I) = A^TA + 2A + I$. Consequently, σ appears as an eigenvalue of A which it necessary might not be correct. In fact, it is true if A is symmetric.

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6. Considering $m \times n$ matrix A with singular decomposition $A = U\Sigma V^T$. What is the singular value decomposition of following matrices?
   (a) A^T.
   (b) AA^T.
   (c) A^TA.

   Answer. Assume that $A \in \mathbb{R}^{m\times n}$ with singular decomposition $A = U\Sigma V^T$ where $U^TU = I$, $V^TV = I$, and Σ is a diagonal matrix containing the singular values of A.

$$A^T = V\Sigma^TU^T.$$

$$AA^T = U\Sigma V^TV\Sigma^TU^T = U\Sigma\Sigma^TU^T.$$

$$A^TA = V\Sigma^TU^TU\Sigma V^T = V\Sigma^T\Sigma V^T.$$
